

Effect of drying environment on mechanical properties, internal RH and pore structure of 3D printed concrete

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ABSTRACT

3D printed concrete (3DPC) is fabricated by depositing printable materials without formwork which increases the surface exposed to environmental condition, and by default, eliminates the curing procedure. As a result, moisture evaporation from printed concrete mitigates the hydration of cement and impacts the development of the mechanical properties. In this study, the effects of drying (RH = 60%±5%), wind (3 m/s) and exposed area on the mechanical properties of 3DPC at 20°C±5°C were investigated. Furtherly, the internal relative humidity evolution was measured by humidity sensor and the pore distribution was evaluated by X-ray computed tomography. Curing condition were found to have a significant influence on compressive and flexural strength development for samples cut from printed elements and cast samples, but had less significant effect on splitting tensile strength. Furthermore, compared with cast specimens, samples cut from printed elements were more sensitive to curing condition. Evidence on the pore structure seemed to explain the difference: the pore connectivity increased and the pore distribution varied for samples cut from printed elements. We also confirmed the existence of anisotropy for samples cut from printed elements and found that it did not aggravate in drying and wind conditions.

1. Introduction

Concrete 3D printing is a new, formwork-free construction technology by which printable materials could be extruded by the nozzle and stacked by successive layers. Compared with traditional concrete casting, concrete printing has the advantages of high automation, excellent construction safety, and time, formwork, and labor cost reduction [1,2]. However, without protection from formwork, 3DPC loses moisture during the printing and after deposition. Environmental factors, such as low relative humidity (RH) and high wind speed, can aggravate the moisture evaporation of the concrete surface [3,4]. Because of the large exposed area of 3DPC, the internal moisture moves to the outside more easily, which hinders the hydration reaction of cement, and induces insufficient hydration products to fill the pores. As a result, mechanical properties of 3DPC decreases due to the increase in porosity and coarse pore [5].

After casting, traditional concrete is protected by formwork, less moisture evaporation can appear at early stage [6]. However, after removing the formwork, concrete is exposed to environmental conditions and its strength development under drying condition (RH = 50%±

5%, T = 20°C±2°C) is lower than that under standard curing condition (RH > 95%, T = 20°C±2°C) [7–9]. Alsayed et al. [10], for example, reported that the compressive strength of concrete cured with sprinkling of water twice a day was 42% higher than that of concrete without curing at 360d. The porosity was reduced by 23.1%, and the water absorption was reduced by 22.5%. The tensile strength of samples cured under drying condition reduced by 30% compared with those cured under standard condition at 70d [11]. Wind affects the mechanical properties as well. Niu [9] found that curing under the wind speeds of 5 m/s and 13 m/s at 20°C and 40% RH reduced the splitting tensile strength by 16% and 44% respectively. The exposure of concrete to environmental conditions results in moisture evaporation from the surface of the concrete, leading to the migration of the water from inside to the surface to complement the water loss on the surface, thus generates the gradient of relative humidity inside the concrete [6]. Because the relative humidity of the surface is relatively low compared with the inside concrete due to the evaporation caused by direct exposure to the ambient, this results in a lack of moisture for hydration reaction in concrete [12,13].

During printing, 3DPC filaments are exposed to drying conditions

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directly after being extruded from the nozzle. As formwork is absent, the moisture loss of 3DPC occurs earlier compared to traditional concrete which has the protection from formwork [14]. In addition to the water evaporation after deposition, the water evaporation during the printing process can also affect the bond properties between two consecutive layers [15]. Moisture evaporation becomes more significant when the time interval between two layers increases, which further reduces interlayer bond strength and mechanical properties of 3DPC [16,17]. Le et al. [18] found that the tensile strength reduces 69.6% when the printing time interval between two layers increases from 0 min to 15 min. Sanjayan et al. [19] estimated that the interlayer bond strength reduces 47.7% as a result of the moisture loss of concrete in drying condition at $23^{\circ}\text{C}\pm 3^{\circ}\text{C}$. The higher stiffness of the printed material and the longer time interval during printing can introduce pores and flaws of printed filaments [20]. Ma et al. [21] confirmed this conclusion from a mesoscopic scale by X-ray computed tomography (CT). The voids and longitudinal defects introduced during the layer deposition can weaken the mechanical properties of concrete under the effect of environment [22].

After deposition, lack of formwork increases the exposed area, which leads to more adverse effects from the environment on mechanical properties [23]. Wolfs et al. [24] found that the flexural strength of 3DPC without covering decreased 26% compared with that of covered samples at 24 h. Keita et al. [25] on the other hand found that the splitting tensile strength of the samples without covering was reduced by 50% when compared with that of covered samples. 3DPC fabricated without formwork can potentially aggravate the effect of environment, as the effect of the environment starts even during the printing. Therefore, it is crucial to study the impact of different curing conditions on mechanical properties of 3DPC.

Former studies observed that mechanical properties development is sensitive to drying [26] and the anisotropy of mechanical properties of 3DPC is attributed to the filaments stacked layer-by-layer [27]. However, little research is conducted on the variation of mechanical properties of 3DPC under the effect of drying condition. Especially, very few studies are conducted with environment controls starting from very early age, i.e., just after printing. Furtherly, the influence of drying and wind on the anisotropy of mechanical properties of 3DPC is still unclear. Therefore, this study aims at characterizing the mechanical properties of samples cut from printed elements under the effect of relative humidity, wind and exposed area. Those mechanical properties are compared with cast samples that follow the same curing condition since their preparation. To explain the difference on those mechanical properties introduced by curing conditions and fabrication methods, the internal RH and pore structure of concrete are characterized.

2. Materials and experiments

2.1. Materials and mix proportions

PII 52.5 Portland cement (PC), was used in this study. Silica fume (SF) was used as supplementary cementitious materials to ensure the printability of 3DPC. The chemical composition of cement and silica fume was determined by X-ray fluorescence spectrometry, as presented in Table 1. The maximum size of sand was 1.18 mm, and the fineness modulus equaled 1.7. Polycarboxylate ether-based superplasticizer (PCE) was incorporated in the mix design to adjust the rheological properties. Hydroxypropyl methylcellulose (HPMC) was used to improve the viscosity and achieve the desired printability. Tap water

was used as mixing water. The basic mix composition of 3DPC is presented in Table 2.

To prepare 3D printable concrete, the Non-Tilting Drum Mixer with the capacity of 60L was used, the mixing speed was 48 r/min. Solid powders (cement, silica fume, sand, PCE and HPMC) were mixed for 1 min first. Water was then added and mixed with the powders for 4 min to obtain a homogenous mixture. The total mixing time was 5 min.

2.2. Material properties

The fresh properties of 3D printable concrete, i.e., flowability, buildability and rheological properties were tested within 10 min after mixing. Flowability was determined by jump table test [28]. The initial and final setting time of cement paste were determined by a manual Vicat device [29]. The buildability was characterized according to the method described in [30], expressed as the height of the sample after demolding from a cylindrical mold whose height and diameter were 80 mm, respectively. A similar method was applied by Zhang, Yu [31]. The yield stress was determined by Brookfield rheometer, using vane spindle with dimension of 20 mm $\times$ 40 mm. The testing protocol was identical to the one indicated in [32], i.e., applying a constant shearing speed of 0.1 s^{-1} for 2 min and the maximum shear stress was obtained as the static yield stress, each test was carried out three times. The fresh properties of mortar are depicted in Table 3. The evolution of static yield stress of 3DPC is shown in Fig. 1.

2.3. Printing parameter

The 3D concrete printer used in this study, manufactured by Nanjing Jiayi Digital Additive Technology Research Institute, had a 3 m $\times$ 3 m $\times$ 3 m frame dimension (Fig. 2a). The diameter of the spherical nozzle was 20 mm. For all the printings, the extrusion speed was fixed at 70 r/min, and the moving speed of the nozzle was 150 mm/s. Because of the round shape of nozzle and the pressure of the nozzle on the filaments, the width of the printed filament was about 25 mm and the height was about 12 mm.

Two printing paths were applied for obtaining samples with different interlayers. As presented in Fig. 2b, Printing path A generates printed elements composed of four vertical layers, and each vertical layer was made of two side-by-side printed filaments. Printing path B allowed to print elements composed of four vertical layers, but each vertical layer was made of seven side-by-side printed filaments.

2.4. Environment control

To study the influence of different environmental conditions (RH, wind and exposed area) on mechanical properties of 3D printed concrete, different curing conditions, as given in Table 4, were considered.

Immediately after deposition, the printed elements were covered with a rectangular PVC box with dimension of 2200 mm (length) $\times$ 1200 mm (width) $\times$ 500 mm (height) for initial curing in given environment. The filaments were printed along the long side of the

Table 2
Mix proportion of 3D printed concrete (kg/m^3).

Cement	SF	Sand	Water	PCE	HPMC
702.7	61.1	1222.0	229.1	1.30	0.38

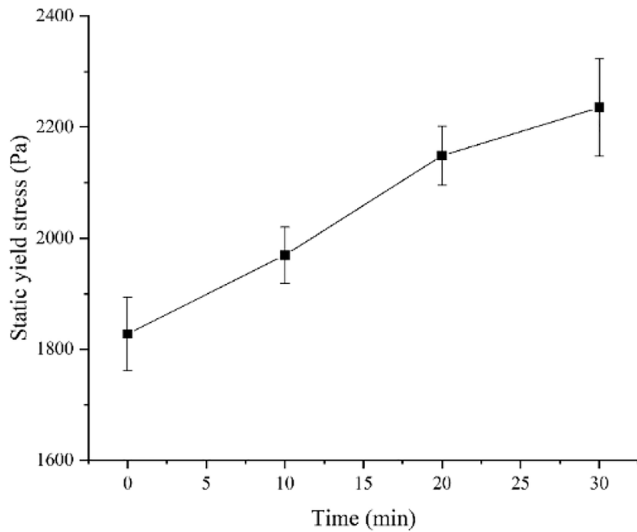
Table 1
Chemical composition of cement and silica fume by XRF (%).

	CaO	SiO ₂	Al ₂ O ₃	Fe ₂ O ₃	MgO	SO ₃	K ₂ O	Na ₂ O	P ₂ O ₅	TiO ₂	LOI
PC	64.55	17.47	3.82	2.96	1.00	3.46	0.65	0.26	0.10	0.21	2.40
SF	–	93.10	0.26	0.06	0.14	0.57	0.22	0.34	0.05	0.01	1.68

Table 3

Fresh properties of 3D printable concrete.

Initial setting time	Final setting time	Buildability	Slump flow
146 min	194 min	75 mm	152 mm

**Fig. 1.** Evolvement of static yield stress of 3D printable concrete.

environmental case, as shown in Fig. 3a. To create different environments inside the box, we used either a humidifier to maintain $RH > 95\%$, or saturated solution of NaBr combined with silica gel was used to maintain $RH = 60 \pm 5\%$. In the beginning, the relative humidity inside the box tended to increase because of the moisture evaporation from fresh concrete, so silica gel was applied to quickly absorb the moisture evaporated from concrete and helped to stabilize RH at this early stage. Inside the box, three ventilators were installed to create an environment with wind speed of 3 m/s in fixed direction (Fig. 3b). The positions of three ventilators were arranged 10 cm away from the printed element to ensure constant and stable wind in the box. When the exposed surface was considered as a factor affecting the properties of 3DPC, the non-exposed surfaces of printed element/samples were covered with plastic film. The cast samples were prepared as well to compare with cut samples from printed elements. Mortar with the sample mix design was prepared following the same procedure as 3DPC. The size of cast specimens for mechanical tests is the same as those of 3DPC specimens. After being cast to molds, the mixtures were vibrated for 2 min.

2.5. Mechanical properties

Immediately after deposition, printed elements were cured in the environmental case for 48 h. Then the samples with sizes of $40 \text{ mm} \times 40 \text{ mm} \times 160 \text{ mm}$ or $40 \text{ mm} \times 40 \text{ mm} \times 40 \text{ mm}$ were cut from printed elements by removing irregular printed surfaces which is normally less than 10 mm thick [33]. It should be mentioned that the surface state of specimens cut from printed elements was slightly different from those of printed elements and cast samples. In drying condition, the moisture on the surface of the filament evaporates first, and then the water inside constantly moves outwards the surface to replenish the lost moisture. Therefore, the printed irregular surface provides the protection for the inner part of printed elements to some extent. However, due to the limited size cut from the printed element, this effect can be ignored. Three groups of samples with different interlayer numbers were obtained, as presented in Fig. 2. To clarify the influence of the environment on the anisotropy of mechanical properties of 3D printed concrete, the samples tested for flexural strength was printed in different ways: for X and Z loading directions, samples contained one vertical interlayer and three horizontal interlayers; and for Y direction loading test, samples cut from printed elements had six vertical interlayer and three horizontal interlayers. For compressive strength and splitting tensile strength, the sample cut from printed elements had one vertical interlayer and three horizontal interlayers. After cutting, the samples were placed either in standard curing condition S, or drying condition D or drying and wind condition W, respectively, as defined in Table 4. The dimension of cast specimens for flexural strength, compressive strength and splitting tensile strength was same as 3DPC. Half of the cast samples were demolded directly, and the rest were wrapped with plastic film for all their surfaces except the top one. All samples (cast and printed samples) were molded in one batch and then stored in specific environment cases until test.

To study the anisotropy of mechanical properties of printed elements, samples cut from printed elements were loaded in X, Y, and Z directions, as shown in Fig. 4. The flexural strength and the compressive strength were obtained in conformity with the Chinese National Standard GB/T 17671-1999 [34]. The splitting tensile strength was tested according to Chinese National Standard GB/T 29417-2012 [35]. All the mechanical tests were conducted at ages of 3d, 7d and 28d. The dimension equalled $40 \text{ mm} \times 40 \text{ mm} \times 160 \text{ mm}$ for flexural strength test

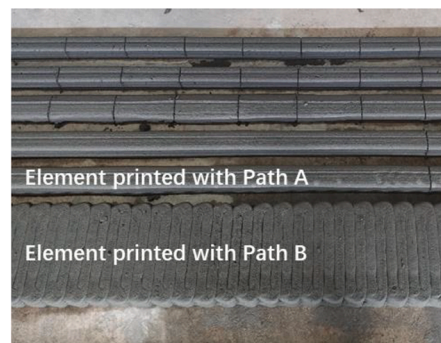
Table 4

Curing conditions.

Curing condition (abbreviation)	Relative humidity (RH)	Temperature	Wind speed
Standard (S)	$>95\%$	$20^\circ\text{C} \pm 5^\circ\text{C}$	0
Drying (D)	$60\% \pm 5\%$	$20^\circ\text{C} \pm 5^\circ\text{C}$	0
Drying and Wind (W)	$60\% \pm 5\%$	$20^\circ\text{C} \pm 5^\circ\text{C}$	3 m/s



(a)



(b)

Fig. 2. (a) 3D printer and (b) printed elements.

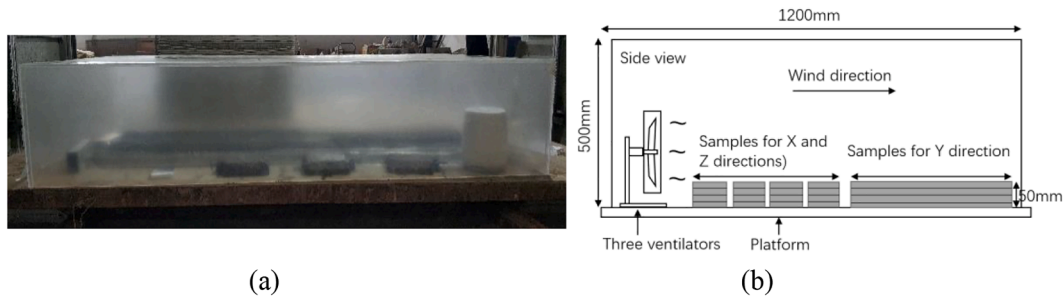


Fig. 3. Curing: (a) environmental case; (b) location of ventilators.

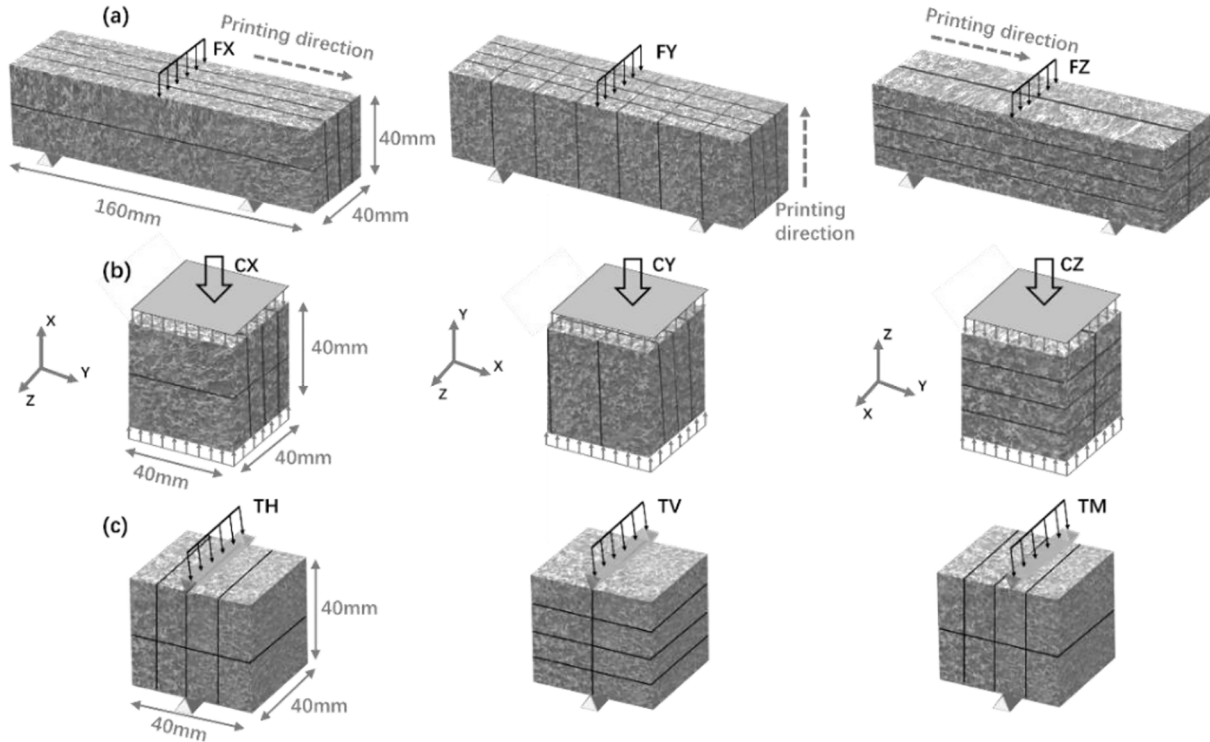


Fig. 4. Loading of samples: (a) flexural; (b) compressive; (c) splitting tensile.

and $40\text{ mm} \times 40\text{ mm} \times 40\text{ mm}$ for compressive and split tensile strength test. The final strength was the average of 3 samples for flexural and split tensile strength, 6 for compressive strength.

2.6. Internal relative humidity

To understand the impact of different curing conditions on the internal RH evolution of 3DPC, humidity sensors were embedded in various samples at different locations to monitor the internal RH change

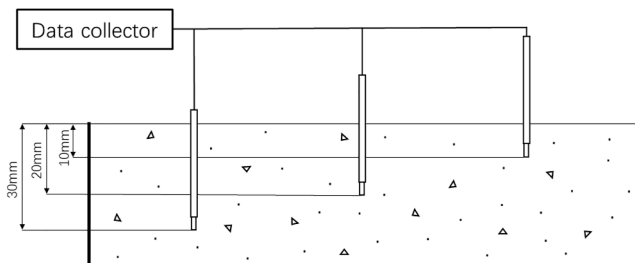


Fig. 5. Sensor locations for the internal RH collection.

continuously, as seen in Fig. 5. The data acquisition was performed by Toprie sensor TP2307 with an accuracy of $\pm 1.8\%$ automatically. The cast samples with a size of $40\text{ mm} \times 40\text{ mm} \times 160\text{ mm}$ with the top surface exposed to different environments were tested at the depth of 10 mm, 20 mm and 30 mm, respectively.

2.7. X-CT to characterize pore structure

The X-ray computed tomography (X-CT) YXLON was used to characterize the pore structure of samples. The sample size for X-CT was $40\text{ mm} \times 40\text{ mm} \times 40\text{ mm}$. The resolution of X-CT images for analysis was 884×884 pixels with a pixel size of $61.8\text{ }\mu\text{m}$. The duration of one CT-scan is 30 min and all the samples tested had the same curing time. The pore threshold was determined by the tangent slope method and was extracted by Dragonfly software as ROI (region of interest) area [36].

Specifically, the cubes with the size of $10\text{ mm} \times 10\text{ mm} \times 10\text{ mm}$ were extracted from different zones of the reconstructed samples for detailed analysis of porosity, the total number of pores, and pore connectivity. Each cube was divided into 161 2D sections, as shown in Fig. 6 (b). The pore structure connectivity analysis was performed based on the results of the pore threshold segmentation, which could establish

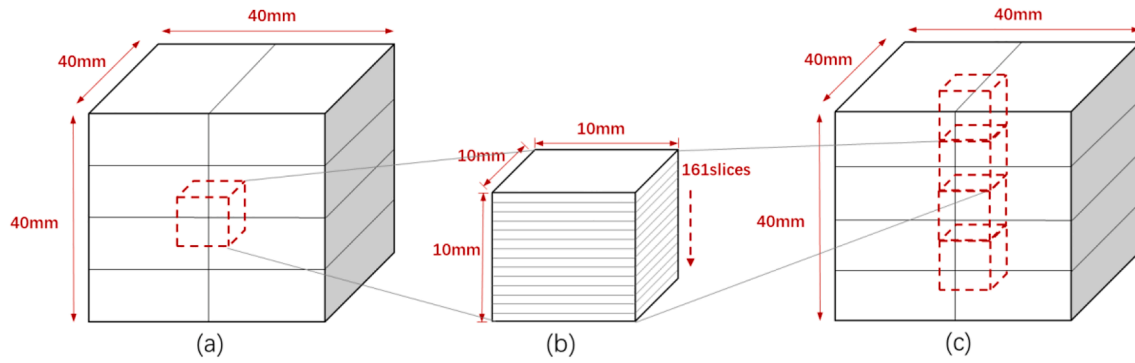


Fig. 6. Sample extracted from the reconstructed 3D structure: (a) locations for analyzing the effect of fabricated method and environment; (b) sampling size and the number of slices for the pore distribution analysis; (c) locations for analyzing the pore distribution at different zones.

spheres and lines representing pores and their connections [37]. Three repetitive samples were measured to calculate the average porosity for each position.

To determine the effect of fabrication methods and curing conditions on the pore distribution, 10 mm cubes involved interface were extracted. Sampling position for pore structure connectivity analysis under the effect of fabrication methods was also located in the center of the sample, as shown in Fig. 6(a). To explore the effect of curing condition on the porosity of printed filament and interlayer from top to bottom of printed sample, 10 mm cubes were extracted from samples, as shown in Fig. 6 (c).

3. Results and discussion

3.1. Effect of curing conditions on compressive strength

The compressive strength development of samples under curing conditions S, D and W, measured at different ages, are compared in Fig. 7. For samples cured under drying and wind condition, all the surfaces were exposed to the curing environment without any protection. When the RH decreased from 95% to 60%, the compressive strength dropped significantly, for both printed and cast samples at all ages. When the samples were cured under W condition, the compressive strength reduced more significantly for all the specimens loaded from different directions. Indeed, the compressive strength decreased significantly under wind condition compared with that in drying condition. The significance on differences between results are checked by t-tests.

The compressive strength reduction rate of samples cured in drying

condition, with or without wind, when compared with those under RH > 95% environment, is showed in Table 5. The compressive strength reduction rate was defined by the difference between the compressive strength in D or W condition and divided by the compressive strength in S condition. From Table 5, one observation was that the compressive strength reduction rate increased with age for all specimens, regardless of preparation method. When comparing the reduction rate between printed and cast specimens in the same curing condition at the same age, the reduction rate of samples cut from printed elements was generally higher than that of cast ones. For the compressive strength, the reduction rate of the printed specimen under drying condition is 13.9%–19% at 3d, and 12.4% for the cast specimens. The reduction further increases to 22–26% for printed specimens and 19.4% for the cast specimen at 7d; At 28d, the value rises to 31–33.3% for the printed specimens and 31.1% for cast specimens, respectively. In dry and wind condition, the compressive strength of printed sample reduces by 14.3–21.9% at 3d, which is a little higher than cast sample of 14.6%. At 7d and 28d, the strength of printed sample reduces to 20–31.8% and 32.2–34.3% respectively in wind condition, which is slightly higher than 22.7% at 7d and 32.2% at 28d for cast samples. In other words, the compressive strength development of samples cut from printed elements was more sensitive to curing condition than that of cast concrete.

3.2. Effect of curing conditions on flexural strength

Fig. 8 represents the flexural strength evolution of printed and cast samples. Consistent with the impact of different curing conditions on the compressive strength, the flexural strength reduced significantly in drying condition, wind aggravated the downward of flexural strength furtherly. The significance on differences between results are checked by t-tests. The flexural strength reduction rate of samples cured in drying condition, with or without wind, when compared with those under RH > 95% environment, is given in Table 6. Comparing the flexural strength

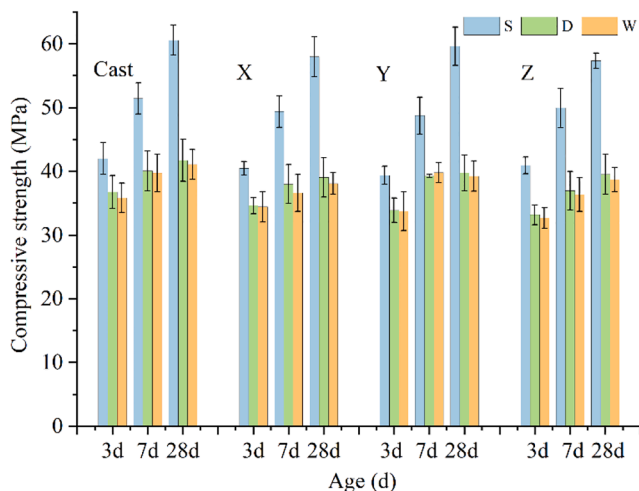


Fig. 7. Effect of curing conditions on compressive strength of printed and cast specimens. X, Y and Z represent the loading directions of 3DPC.

Table 5

The compressive strength reduction rate of specimens cured under D and W condition when compared with S condition.

Curing condition	Age	D condition (%)	W condition (%)
Printed, loaded at X direction	3d	14.5 ± 0.07	15.0 ± 0.10
	7d	23.1 ± 0.06	25.9 ± 0.07
	28d	32.6 ± 0.09	34.3 ± 0.08
Printed, loaded at Y direction	3d	13.9 ± 0.05	14.3 ± 0.07
	7d	22.0 ± 0.09	20.0 ± 0.09
	28d	33.3 ± 0.10	34.2 ± 0.07
Printed, loaded at Z direction	3d	19.0 ± 0.09	21.9 ± 0.09
	7d	26.0 ± 0.09	31.8 ± 0.09
	28d	31.0 ± 0.09	32.2 ± 0.07
Cast	3d	12.4 ± 0.06	14.6 ± 0.06
	7d	19.4 ± 0.10	22.7 ± 0.10
	28d	31.1 ± 0.08	32.2 ± 0.05

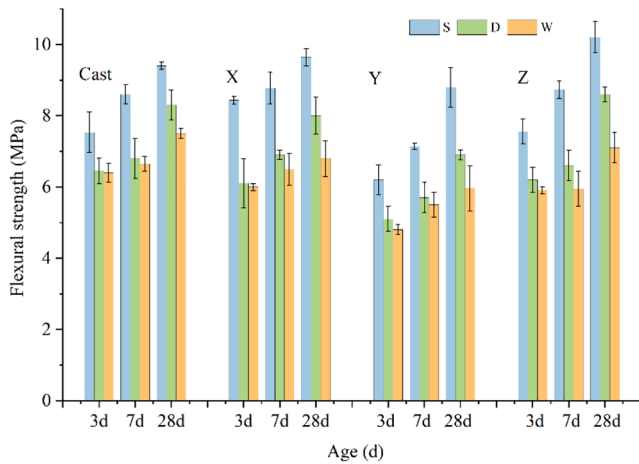


Fig. 8. Effect of curing conditions on flexural strength of printed and cast samples. X, Y, and Z represent the loading directions of 3DPC.

Table 6

The flexural strength reduction rate of specimens cured under D and W condition when compared with S condition.

Curing condition	Age	D condition (%)	W condition (%)
Printed, loaded at X direction	3d	27.7 ± 0.10	28.9 ± 0.07
	7d	21.4 ± 0.08	25.9 ± 0.07
	28d	17.0 ± 0.07	29.5 ± 0.12
Printed, loaded at Y direction	3d	17.7 ± 0.11	22.6 ± 0.02
	7d	20.2 ± 0.05	23.0 ± 0.09
	28d	21.6 ± 0.07	32.2 ± 0.08
Printed, loaded at Z direction	3d	18.5 ± 0.09	20.1 ± 0.09
	7d	24.4 ± 0.09	27.2 ± 0.04
	28d	16.2 ± 0.05	32.5 ± 0.02
Cast	3d	14.4 ± 0.07	15.0 ± 0.05
	7d	20.9 ± 0.07	22.7 ± 0.09
	28d	11.7 ± 0.05	20.2 ± 0.05

reduction rate between samples cut from printed elements and cast samples in the same curing condition at the same age, the flexural strength reduction rate of samples cut from printed elements was in general higher than that of cast ones. This indicated again that the mechanical properties of samples cut from printed elements were more sensitive to environment when compared with cast samples.

From Fig. 8, the flexural strength of samples cut from printed elements varied with loading directions: the flexural strength loaded at Y direction was lower than that of X and Z direction at 28d. It can be seen from Fig. 4 that for printed samples loaded at Y direction, there were 6 interfaces parallel to the loading direction, which was the most unfavorable loading direction. Consequently, the flexural strength loaded at Y direction was the lowest. The flexural strengths of samples cut from printed elements loaded at X and Z directions were similar or slightly higher than that of cast samples. From Fig. 4, we can find that there was no interlayer parallel to the loading direction X or Z. In addition, the extrusion process may densify the concrete at these two directions [18].

3.3. Effect of environment on compressive and flexural strength

From Fig. 7, the compressive strength of printed concrete was lower than that of cast samples, and no obvious anisotropy was observed for compressive strengths in different curing conditions. Under the effect of drying and wind, as shown in Fig. 8, the flexural strength of samples loaded at Y direction was lower than the other two directions. But the flexural strength reduction rate for samples loaded at Y direction was similar to that of the other two directions as can be observed from Table 6, i.e., under the drying condition, the flexural strength reduction rate on Y direction equaled 17.7% at 3d, and it was 27.8% and 18.5% for

X and Z directions, respectively. Y direction was 20.2% at 7d, and it was 21.4% and 24.4% for X and Z directions, respectively. Y direction was 21.6% at 28d, and it was 17.0% and 16.2% for X and Z directions, respectively. This indicated that drying and wind slightly affect the anisotropy of mechanical properties in case of 3DPC.

To further compare the impact of different curing conditions on the mechanical anisotropy of 3D printed concrete, we refer to the mechanical anisotropy index (I_a), proposed by Junhong Ye [38]:

$$I_{Direction} = \sqrt{(f_x - f_{Direction})^2 + (f_y - f_{Direction})^2 + (f_z - f_{Direction})^2} / f_{Direction} \quad (1)$$

where f_x , f_y , f_z [MPa] are the mechanical properties loaded at X, Y and Z directions, respectively. The footnote 'Direction' is one of the loading directions. When the loading direction is X, the f_x represents the strength loaded at X direction and I_x represents the anisotropic index at X direction. The lower the mechanical anisotropy index, the less significant the mechanical anisotropy is.

Table 7 presents the compressive anisotropy index in S, D and W curing conditions. It can be observed that the anisotropy indexes of compressive strength were all very low, therefore no obvious compressive anisotropy exists for the printed concrete investigated in this study. The anisotropy of 3D printed concrete is related to the defects of interface and the variation of pore distribution [39]. When printing parameters match well with the printable material, the printed elements will have less interface defects, and the anisotropy of mechanical properties becomes less evident [40,41].

Table 8 shows the anisotropy index of flexural strength under S, D and W curing conditions. The anisotropy index of flexural strength at 28d was lower than that at 3d and 7d in three curing conditions, especially for Y direction. The mechanical anisotropy is caused by the existence of interlayers between printed layers which are normally weak than bulk layers. The improved bond properties between interlayers are attributed to the densification of the microstructure caused by continuous hydration of cement at later ages [42]. Thus, the anisotropy of flexural strength mitigates.

From Table 8, it can also be seen that the anisotropy of flexural strength at the same age is similar in the three environments, with a maximum difference of 0.08. Because of the small sample sizes and the different variance, we applied *t*-test to clarify the significance of the difference. The result shows that the P-value of the anisotropy index under different environments is higher than 0.05, which means the difference between anisotropy indexes obtained from flexural strengths measured in standard, drying, and wind conditions is not significant. Thus, we concluded that the anisotropy of flexural strength of 3D printed concrete is not aggravated by curing environment.

Table 7

Anisotropy index of compressive strength for printed sample under different curing conditions.

X direction (Ix)			
Curing time	S condition	D condition	W condition
3d	0.03	0.05	0.05
7d	0.02	0.04	0.06
28d	0.03	0.02	0.03
Y direction (Iy)			
Curing time	S condition	D condition	W condition
3d	0.05	0.03	0.04
7d	0.03	0.07	0.08
28d	0.05	0.02	0.03
Z direction (Iz)			
Curing time	S condition	D condition	W condition
3d	0.04	0.05	0.06
7d	0.03	0.07	0.07
28d	0.04	0.01	0.02

Table 8

Anisotropy index of flexural strength for printed sample under different curing conditions.

X direction (I_x)	S condition	D condition	W condition
Curing time			
3d	0.28	0.27	0.24
7d	0.22	0.20	0.22
28d	0.19	0.21	0.20
Y direction (I_y)			
Curing time	S condition	D condition	W condition
3d	0.42	0.34	0.38
7d	0.29	0.29	0.32
28d	0.27	0.25	0.24
Z direction (I_z)			
Curing time	S condition	D condition	W condition
3d	0.23	0.29	0.22
7d	0.22	0.22	0.24
28d	0.20	0.21	0.22

3.4. Effect of exposed areas on mechanical properties of 3DPC cured in 60% RH

Fig 9 presents the compressive strength and flexural strength at 28d in function of exposed area for samples cut from printed elements and cast samples. When the exposed area equals to 0, it means no drying happens during curing, i.e., samples are cured in standard condition. $6.4 \times 10^{-3} \text{m}^2$ represents top surface exposure and $28.8 \times 10^{-3} \text{m}^2$ represents all surfaces exposure. As similar tendency on the compressive and flexural strength evolution at a given age, i.e., at 3d, 7d and 28d in function of exposure was found, only the correlation between the exposure area and the 28d strengths was plotted and analyzed. The strength of the printed sample and cast sample decreases markedly when one surface was exposed to drying curing. But the further reduction becomes less significant when all the surfaces are exposed compared with that of one surface exposure.

For the anisotropic properties of the specimen on compressive and flexural strength, the compressive strength loaded at X, Y, Z directions are similar (Fig. 9a). The flexural strength loaded at Y direction is lower than that of X and Z directions in three different exposed areas (Fig. 9b), indicating that the anisotropic properties exist in flexural strength no matter how many surfaces are exposed to drying conditions.

The mechanical properties reduction rate of specimens exposed compared with that of not exposed is shown in Table 9. For the flexural strength, reduction rate of printed samples was higher than that of cast samples. For the compressive strength, the reduction rate of the printed samples was slightly higher compared with cast samples with a few exceptions. This further confirms that 3D printed elements are more sensitive to surface exposure and curing environment than conventional

Table 9

Strength reduction rates of samples cut from printed elements and cast samples with different exposed areas in drying condition.

	Exposure condition	Age	Cast	Printed X direction	Printed Y direction	Printed Z direction
Flexural strength reduction	One surface exposure	3d	4.4 ± 0.09	13.8 ± 0.08	11.3 ± 0.15	11.7 ± 0.07
		7d	11.2 ± 0.05	11.9 ± 0.08	12.7 ± 0.04	15.2 ± 0.06
		28d	7.8 ± 0.09	7.7 ± 0.04	14.2 ± 0.11	13.7 ± 0.05
	All surfaces exposure	3d	14.4 ± 0.09	27.7 ± 0.11	17.7 ± 0.10	18.5 ± 0.07
		7d	20.9 ± 0.09	21.4 ± 0.05	20.2 ± 0.08	24.4 ± 0.07
		28d	11.7 ± 0.05	17.0 ± 0.07	21.6 ± 0.07	16.2 ± 0.05
	One surface exposure	3d	7.9 ± 0.07	4.8 ± 0.05	7.7 ± 0.08	9.4 ± 0.07
		7d	19.2 ± 0.07	23.1 ± 0.09	17.9 ± 0.08	21.2 ± 0.10
		28d	26.4 ± 0.07	28.2 ± 0.08	26.3 ± 0.06	25.3 ± 0.08
	All sides exposure	3d	12.4 ± 0.09	14.5 ± 0.05	13.9 ± 0.07	19.0 ± 0.06
		7d	19.4 ± 0.09	23.1 ± 0.09	22.0 ± 0.06	26.0 ± 0.10
		28d	31.1 ± 0.09	32.6 ± 0.10	33.3 ± 0.09	31.0 ± 0.08

cast concrete.

3.5. Effect of curing environment on splitting tensile strength

Fig. 10 (b) shows the splitting tensile strength of samples cut from printed elements under S, D and W curing conditions at 3d, 7d, and 28d. H, V and M denote loading along the horizontal interlayer, the vertical interlayer, and the matrix, respectively (Fig. 10a). As can be expected, the splitting tensile strength loaded at the matrix was much higher than those loaded along the interfaces of 3DPC, with the SM group being

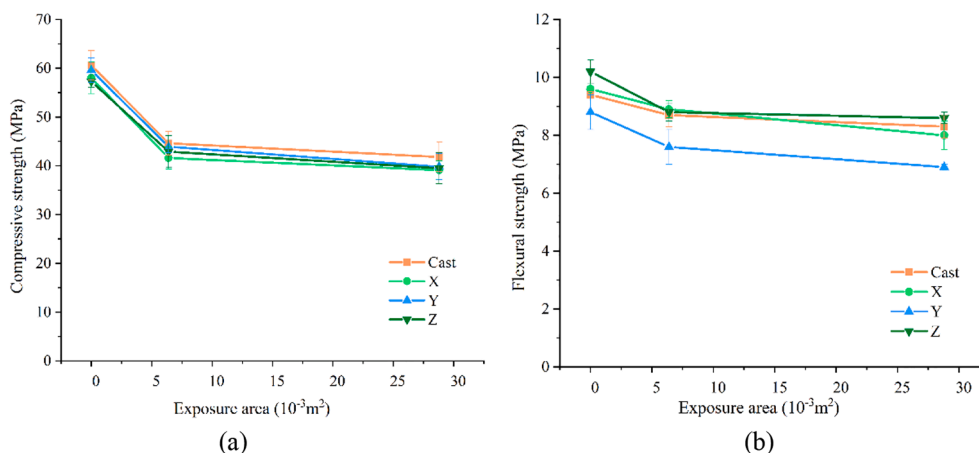


Fig. 9. Effect of exposed area on mechanical properties of printed and cast sample: (a) compressive strength at 28d; (b) flexural strength at 28d.

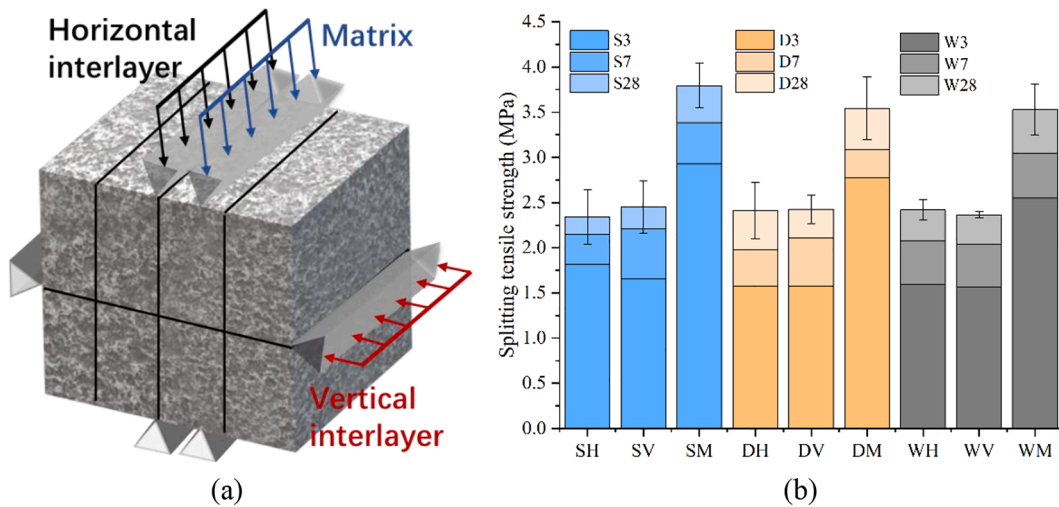


Fig. 10. Splitting tensile strength of different curing conditions: (a) loading positions: horizontal interlayer (H), vertical interlayer (V) and matrix (M); (b) splitting tensile strength under S, D and W curing conditions at 3d, 7d and 28d.

slightly higher than the other two. When loaded along either the horizontal or the vertical interfaces, the splitting tensile strength was in fact very close regardless of the curing conditions. Therefore, curing condition did not affect the splitting tensile strength of 3DPC at 28d in a significant way, although the splitting tensile strength of 3DPC cured under D and W conditions at 3d and 7d was slightly lower than that cured under S condition. As defined in Fig. 10 (a), the splitting tensile force was directly loaded along the interfaces formed during 3D printing process. Therefore, the bonding between adjacent printed filaments, other than curing condition, was the key parameter determining the tensile capacity of the interfaces. The reason for the slight effect of different curing conditions on splitting tensile strength can be explained by the pore structure which will be discussed in section 3.8.

3.6. Internal RH evolution under different curing conditions

The results above confirm that drying and wind both reduce the mechanical properties (compressive, flexural and splitting tensile strength) of 3DPC at given ages, which is consistent with the trend observed for cast samples. However, the flexural and compressive strength reduction rate of samples cut from printed elements was higher than that of cast samples under drying and wind curing. In other words, 3DPC was more sensitive to environment. Furthermore, the environment slightly affected the splitting tensile strength of interlayers and flexural strength anisotropy of 3DPC. To explain this tendency observed on mechanical properties, the internal relative humidity evolution at different depths inside samples was measured, and the X-CT was used to analyze the pore distribution of samples in section 3.7.

The internal RH evolution of the samples with top surface exposed under S, D and W curing conditions is shown in Fig. 11. When cured under standard condition, the internal relative humidity of sample remained constant till 28d. However, when $RH = 60\% \pm 5\%$, the internal RH decreased continuously, as shown by the black lines in Fig. 11. When cured under drying and wind, in the first 8d, the internal RH dropped rapidly, as shown by orange lines in Fig. 11, the sample lost moisture at a higher rate compared with under drying condition, and RH declined continuously until 28d.

D and W curing resulted in humidity gradient inside the samples, with the highest RH appeared in the inmost of sample. It was reported that when the internal RH is lower than 80%, the hydration of cement ceases [43,44]. In fact, at the depth of 10 mm, the date when RH dropped below 80% occurred on the 8th day in W condition and the 10th day in D condition. At the depth of 20 mm, the date occurred on the 17th day in W condition and the 28th day in D condition; and at the depth of

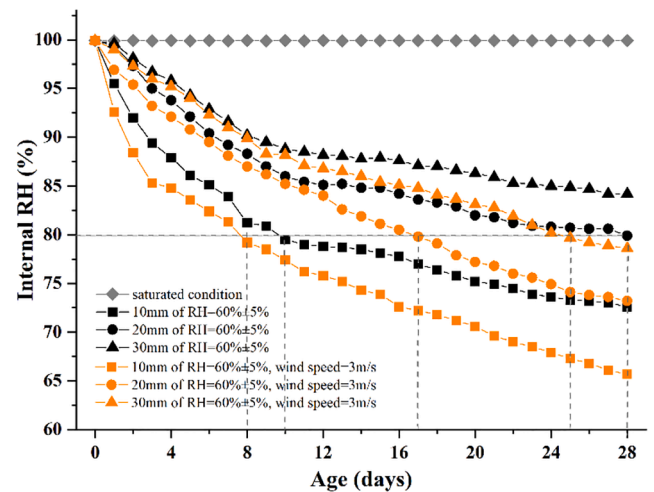


Fig. 11. Evolution of internal relative humidity with time at the depth of 10 mm, 20 mm, and 30 mm from upper surface of sample cured under different conditions.

30 mm, it was on the 25th day in W condition. The internal RH in D condition was above 80% until 28d at the depth of 30 mm. As the depth increased, the effect of environment reduced, and the cementitious material could hydrate continuously. In plastic stage, humidity gradient forms from surface to inside of concrete under drying condition, the drying phenomenon and even plastic cracking appears on the surface of concrete [45,46]. At hardened stage, the loss of moisture inside concrete leads to insufficient hydration of cement and induces more pores, resulting in the decrease of mechanical properties [47,48].

The compressive strength of samples cut from printed elements as a function of internal relative humidity at depth of 20 mm in three loading directions is shown in Fig. 12. The red lines are fitted results showing the linear relation between RH and compressive strength. The relative humidity on each red line represents W, D, S conditions from small to large. It can be seen that the compressive strength was linearly related to the internal RH at specific ages regardless of the loading direction, that is, increasing the internal RH enhanced the compressive strength. By comparing the slopes of the fitted lines, it can be found that the slope at 3d and 7d varied from 0.54 to 0.82 and 0.80 to 0.98, respectively. However, at 28d the slope became narrowed between 0.63 and 0.68. It can be further noted that the slope at 7d was the highest, suggesting that

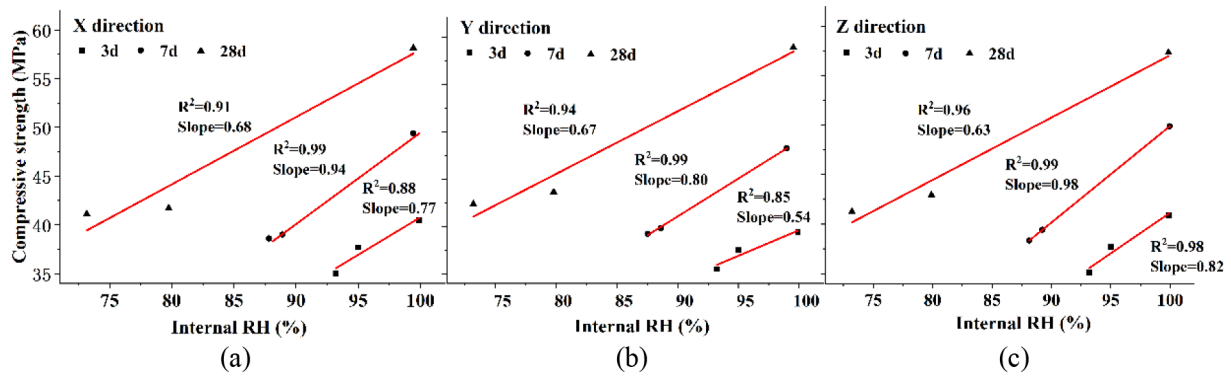


Fig. 12. Correlation between compressive strength and internal relative humidity under S, D and W curing conditions loaded at (a) X direction; (b) Y direction; (c) Z direction.

curing still played an important role on the strength development of 3DPC at 7d. These findings show that the early curing condition is significant to the strength development of 3DPC, for which early curing is often not available, especially in the case of on-site 3D printing.

3.7. Pore structure of printed elements and cast samples

X-CT was used to characterize pore structure after 28d of hardening. The pore structures of printed and cast samples cured in standard condition at 28d are shown in Fig. 13. It can be observed that the pores of the printed sample along the printing direction are elongated. The main reason was associated with the extrusion process during printing and the forces applied on printed layer when stacking new layer above. Similar phenomenon was reported by Panda et al. [49]. And the visible interfaces existed between two adjacent printed filaments, as can be seen from Fig. 13(a). In addition, the pressure during extrusion can compact the concrete and result in smaller pores compared with the cast concrete as shown in Fig. 13(b). A similar observation was found by Kruger [50,51].

Fig. 14 presents the number of pores and porosity as a function of pore size for both samples cut from printed elements and cast samples cured in standard condition at 28d. The results were obtained based on the statistics of 10 mm × 10 mm samples and the printed sample contained a horizontal interface. The porosity of samples cut from printed elements is 4.4%, and the pore size is concentrated in the range of 0.1 mm-0.8 mm with the critical pore size being between 0.4 mm and 0.6 mm (Fig. 14a). In comparison, the porosity of the cast sample is 5.2%, and the critical pore size is similar to the printed sample (0.4 mm-0.6 mm) (Fig. 14b). It can be seen that the number of pores at the range of

0.1 mm-0.4 mm for the printed sample is significantly higher compared with the cast sample. Studies have shown that pores with size > 0.1 mm could negatively affect the concrete strength [52]. The smallest pore size that X-CT can detect in this study was 61.8 μm, which meant all the increased pore number of samples cut from printed elements had negative effect on the mechanical properties. In addition, the maximum pore size of the printed sample increases to 2.2 mm-2.4 mm due to the interlayer phenomenon of the printed sample. The values obtained agreed well with the research by Chen Y [15].

To compare the pore connectivity between samples cut from printed elements and cast samples, the reconstructed cube of 10 mm × 10 mm × 10 mm was extracted at the center of 40 mm cubic specimen cured in S condition at 28d, and the results are shown in Fig. 15. It is evident that sample cut from 3D printed elements has higher connected pore structure than cast sample. The maximum connected length for 3D printed sample equals 0.61 mm, while it is 0.34 mm for cast sample. It is possible that the increased number of pores between 0.1 mm and 0.4 mm in printed sample enhanced the pore connectivity. The connectivity enhancement accelerated moisture migration to outside in drying condition, and the hydration degree of concrete thus decreased, resulting in the reduction of mechanical properties. Fig. 16.

3.8. Effect of curing conditions on pore distribution

Fig 16 displays the effect of drying and wind on the number of pores, pore size and porosity of samples cut from printed elements cured in D and W conditions at 28d. Small cube with dimension of 10 mm × 10 mm × 10 mm was cut from the central position of printed sample. The porosity of 3DPC in D and W conditions is 5.7% and 6.3%, respectively,

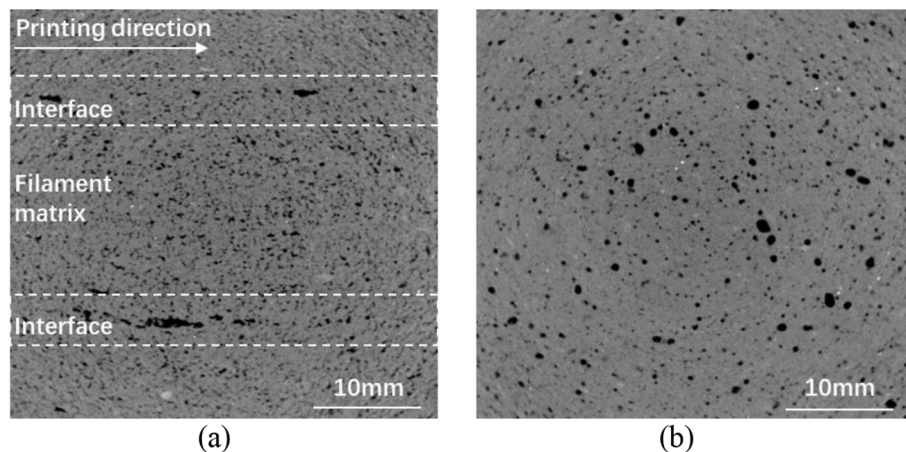


Fig. 13. Effect of fabrication method on pore structure of concrete reconstructed by X-CT: (a) printed sample; (b) cast sample.

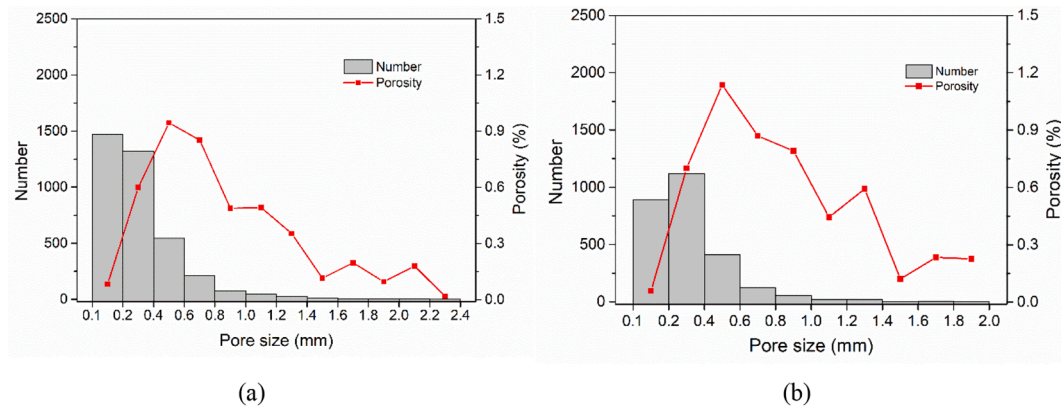


Fig. 14. The number of pores and pore size distribution under two fabrication methods: (a) printed sample; (b) cast sample.

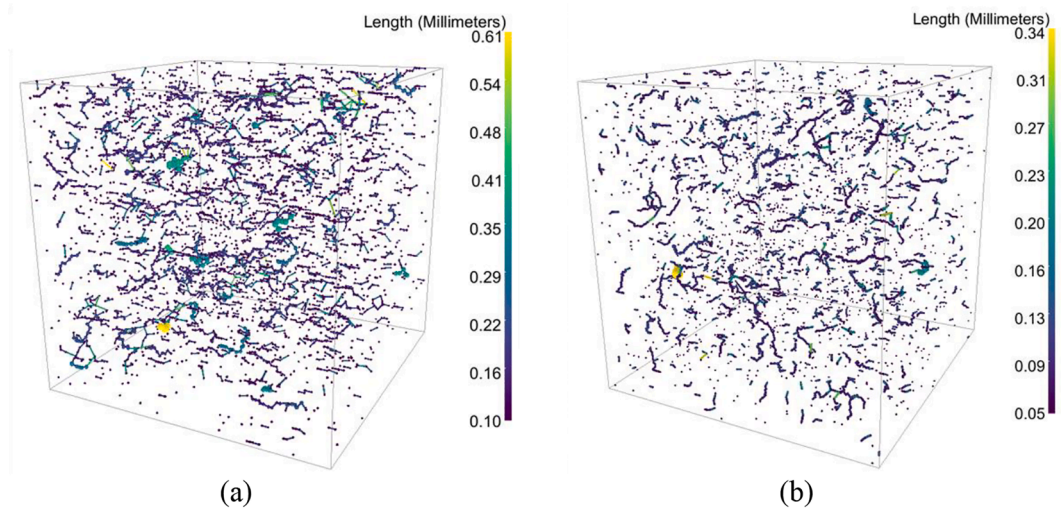


Fig. 15. Pore connectivity of (a) printed sample; (b) cast sample.

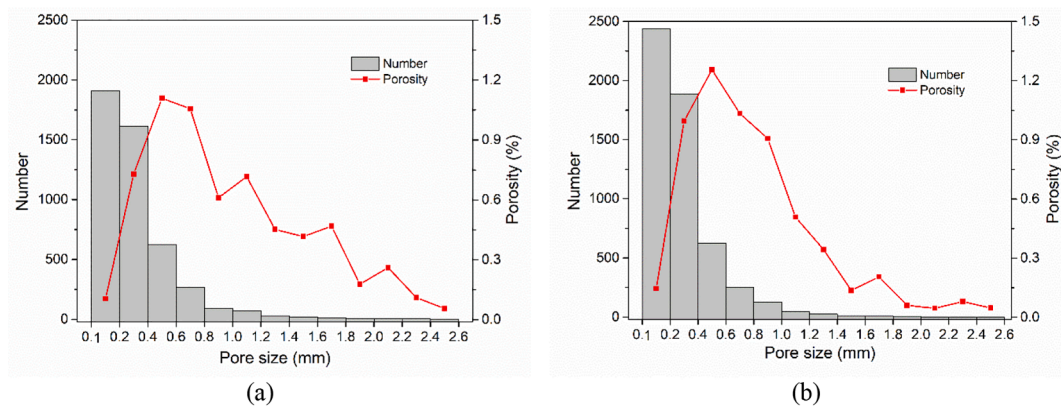


Fig. 16. Effect of curing environments on pore distribution: (a) D condition; (b) W condition.

while it is only 4.4% in S condition. Consistent with standard curing condition, the critical pore size in drying condition and wind condition is 0.4 mm-0.6 mm. In addition, the number of pores at the range of 0.1 mm-0.4 mm increases in wind condition, which further explains the reduction in mechanical strength of 3DPC under wind curing environment.

From section 3.6, we found that the internal RH gradient occurs in samples cured under D and W conditions. To explore the effect of drying

and wind on the porosity at various zones, the 10 mm × 10 mm × 10 mm reconstructed samples were extracted from top filament (TF), top interlayer (TI), bottom interlayer (BI) and bottom filament (BF) of printed sample separately for analysis, as seen in Fig. 17(a) and (b).

As can be observed from Fig. 17(c), the porosity decreased from the top to the bottom of 3DPC. On the one hand, this can be explained by the densification of bottom layers imposed by the weight from upper layers. On the other hand, it can also be seen from Fig. 17(c) that drying and

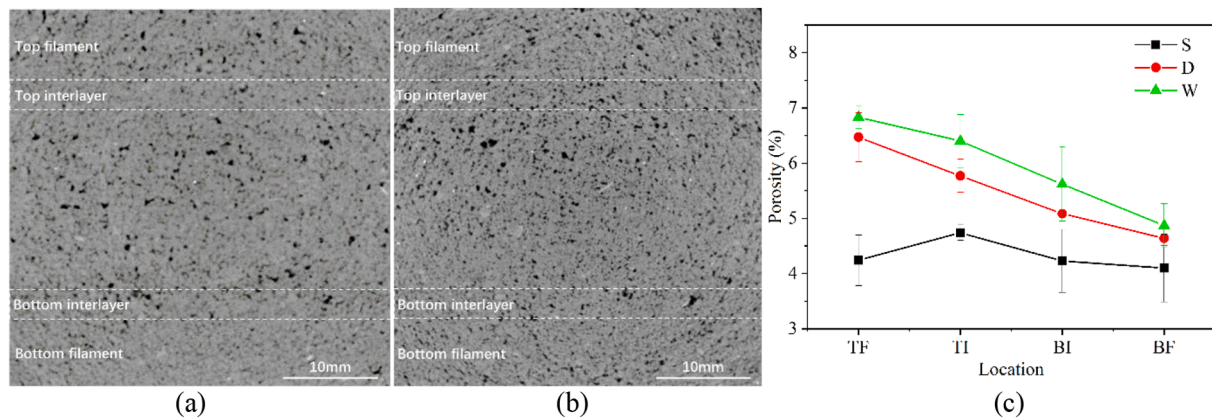


Fig. 17. Effect of environments on porosity: (a) pore distribution in D condition; (b) pore distribution in W condition; (c) porosity of printed sample under S, D and W curing condition.

wind curing caused the increase of porosity in the whole sample due to the moisture loss in concrete, resulting in insufficient cement hydration and more preferential ingress paths. Since the internal RH dropped markedly from the inside to the top of concrete (section 3.6), it was reasonable that the top part of samples had the highest porosity. Therefore, the environment has a slight effect on the inside of concrete. This also explains why the impact of drying and wind on the splitting tensile strength along the interface and the mechanical anisotropy of 3DPC could be neglected.

4. Conclusions

3D printed concrete is fabricated without formwork and exposed to the environmental conditions after printing, therefore this research work focused on the impact of environment on mechanical properties of 3DPC. The following conclusions can be addressed:

(1) The strength of printed sample in drying condition was significantly lower compared with samples cured in standard condition (RH > 95%). The flexural and compressive strength decreased by 20%–30% under 60%±5% RH. Wind (air velocity = 3 m/s) further slows down strength development, the flexural and compressive strength decreased more than 30% under drying and wind curing. However, the drying and wind curing conditions had slight effect on the splitting tensile strength, since the interior of 3DPC was not evidently influenced by curing condition.

(2) In drying condition (RH = 60%±5%), the internal relative humidity decreased with time, and the wind (wind speed = 3 m/s, RH = 60%±5%) further aggravated the decrease of internal relative humidity. The mechanical properties at specific ages were linearly correlated to internal relative humidity of 3DPC.

(3) The severe curing condition, i.e., lower RH, higher exposure area, wind, etc. did not show significant effect on anisotropy of 3DPC, because the dominant factor determining the anisotropy of 3DPC is the properties of interlayer between printed filaments which is not significantly affected by curing condition.

(4) Compared with cast samples, the strength of samples cut from printed elements was more sensitive to environment. The main reasons are associated with the fabricating process. On the one hand, the pore shape of the printed sample tends to be elongated after extrusion and deposition of the filaments; Additionally, the vibration-free process leads to an increase in the number of pores with a diameter larger than 0.1 mm, thus pore connectivity is enhanced.

CRedit authorship contribution statement

Lei Ma: Conceptualization, Investigation, Writing – original draft, Formal analysis, Data curation. **Qing Zhang:** Conceptualization,

Methodology, Writing – review & editing, Supervision, Funding acquisition. **Zijian Jia:** Conceptualization, Methodology. **Chao Liu:** Methodology. **Zhicong Deng:** Methodology. **Yamei Zhang:** Conceptualization, Methodology, Writing – review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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